

Worksheet — Laws of Motion (Class 12 NIOS Physics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions. Show every step in subjective questions; choose one option for MCQs. Take $g = 10 \text{ m/s}^2$ unless otherwise stated. Total time: 50 minutes.

Q1. 1 newton equals: [1 mark]

- (A) $1 \text{ kg} \cdot \text{m} \cdot \text{s}$
- (B) $1 \text{ kg} \cdot \text{m/s}$
- (C) $1 \text{ kg} \cdot \text{m/s}^2$
- (D) $1 \text{ kg} \cdot \text{m}^2/\text{s}^2$

Q2. Action and reaction forces act on the same body. [1 mark]

- (A) True
- (B) False

Q3. Linear momentum is defined as $p = \underline{\hspace{1cm}}$. [1 mark]

Q4. A 4-kg body is acted on by a 16-N net force. Its acceleration is: [2 marks]

- (A) 2 m/s^2
- (B) 4 m/s^2
- (C) 8 m/s^2
- (D) 16 m/s^2

Q5. A 5-kg block on a horizontal surface ($\mu_k = 0.2$) is pushed by a 20-N horizontal force. Its acceleration is: [2 marks]

- (A) 1 m/s^2
- (B) 2 m/s^2
- (C) 3 m/s^2
- (D) 4 m/s^2

Q6. A 0.1-kg ball moving at 20 m/s is caught by a fielder, coming to rest in 0.05 s. Find (i) the impulse and (ii) the average force on the ball. [3 marks]

Q7. A 2-kg gun fires a 0.005-kg bullet horizontally at 500 m/s. Find the recoil velocity of the gun. [3 marks]

Q8. Two blocks of masses 6 kg and 4 kg hang on either side of a frictionless pulley over an inextensible rope. Find the acceleration of the system and the tension. [4 marks]

Q9. A 5-kg block on an inclined plane at 30° with $\mu_k = 0.2$ ($g = 10 \text{ m/s}^2$). (a) Draw the FBD and write equations along and perpendicular to the incline. (b) Find the acceleration of the block down the slope. [5 marks]

Q10. (a) State Newton's three laws of motion. (b) Use Newton's third law to derive the conservation of linear momentum for two interacting particles in an isolated system. (c) A 3-kg ball moving at 8 m/s collides head-on with a stationary 1-kg ball; they stick together after collision. Find the common velocity afterwards. [8 marks]

Answer Key

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Q1. 1 newton equals:

Answer: $1 \text{ kg} \cdot \text{m/s}^2$

$F = ma \rightarrow \text{kg} \times \text{m/s}^2$.

MCQ · EASY · 12-NIOS-PHY-LM-01

Q2. Action and reaction forces act on the same body.

Answer: False

They act on different bodies — that's why they don't cancel.

TRUE_FALSE · EASY · 12-NIOS-PHY-LM-04

Q3. Linear momentum is defined as $p = \underline{\hspace{1cm}}$.

Answer: mv

Mass $\times$ velocity (vector).

FILL_BLANK · EASY · 12-NIOS-PHY-LM-03

Q4. A 4-kg body is acted on by a 16-N net force. Its acceleration is:

Answer: 4 m/s^2

$a = F / m = 16/4 = 4 \text{ m/s}^2$.

MCQ · EASY · 12-NIOS-PHY-LM-03

Q5. A 5-kg block on a horizontal surface ($\mu_k = 0.2$) is pushed by a 20-N horizontal force. Its acceleration is:

Answer: 2 m/s^2

$f_k = \mu_k m g = 0.2 \times 5 \times 10 = 10 \text{ N}$; net = $20 - 10 = 10 \text{ N}$; $a = 10/5 = 2 \text{ m/s}^2$.

MCQ · MEDIUM · 12-NIOS-PHY-LM-07

Q6. A 0.1-kg ball moving at 20 m/s is caught by a fielder, coming to rest in 0.05 s. Find (i) the impulse and (ii) the average force on the ball.

**Answer: $\Delta p = m(v_f - v_i) = 0.1(0 - 20) = -2 \text{ kg} \cdot \text{m/s}$. So |Impulse| = $2 \text{ N} \cdot \text{s}$, opposite to motion.
Average force $F = \Delta p / \Delta t = -2/0.05 = -40 \text{ N}$. Magnitude 40 N opposing motion.**

Impulse–momentum theorem.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-03

Q7. A 2-kg gun fires a 0.005-kg bullet horizontally at 500 m/s. Find the recoil velocity of the gun.

Answer: Momentum conservation (initial = 0): $0 = (2)(v_g) + (0.005)(500)$. So $2 v_g = -2.5 \rightarrow v_g = -1.25 \text{ m/s}$. The gun recoils at 1.25 m/s opposite to the bullet's direction.

Conservation of linear momentum.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-05

Q8. Two blocks of masses 6 kg and 4 kg hang on either side of a frictionless pulley over an inextensible rope. Find the acceleration of the system and the tension.

Answer: $a = (m_1 - m_2)g/(m_1 + m_2) = (6 - 4)(10)/(6 + 4) = 20/10 = 2 \text{ m/s}^2$. $T = 2 m_1 m_2 g/(m_1 + m_2) = 2(6)(4)(10)/10 = 48 \text{ N}$.

Standard Atwood's machine formulas.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-08

Q9. A 5-kg block on an inclined plane at 30° with $\mu_k = 0.2$ ($g = 10 \text{ m/s}^2$). (a) Draw the FBD and write equations along and perpendicular to the incline. (b) Find the acceleration of the block down the slope.

Answer: (a) Forces: weight $mg = 50 \text{ N}$ (vertical down); normal N (perpendicular to incline); friction f_k (up the slope, since the block slides down). Components: along the incline (down-slope = +): $mg \sin \theta - f_k = m a \rightarrow 25 - f_k = 5 a$. Perpendicular: $N - mg \cos \theta = 0 \rightarrow N = mg \cos \theta = 50 \times \cos 30^\circ = 50 \times (\sqrt{3}/2) \approx 43.30 \text{ N}$. (b) Kinetic friction $f_k = \mu_k N = 0.2 \times 43.30 \approx 8.66 \text{ N}$. So $25 - 8.66 = 5 a \rightarrow a = 16.34/5 \approx 3.27 \text{ m/s}^2$ down the slope. Equivalently $a = g(\sin \theta - \mu_k \cos \theta) = 10 \times (0.5 - 0.2 \times 0.866) = 10 \times 0.327 = 3.27 \text{ m/s}^2$.

Standard inclined-plane analysis.

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Q10. (a) State Newton's three laws of motion. (b) Use Newton's third law to derive the conservation of linear momentum for two interacting particles in an isolated system. (c) A 3-kg ball moving at 8 m/s collides head-on with a stationary 1-kg ball; they stick together after collision. Find the common velocity afterwards.

Answer: (a) 1st: A body continues in rest or uniform motion in a straight line unless acted on by a net external force. 2nd: $F = dp/dt$; for constant mass $F = ma$. 3rd: To every action there is an equal-and-opposite reaction (acting on a different body). (b) Take two particles of masses m_1 and m_2 interacting only with each other. By Newton's 3rd law, $F_{12} = -F_{21}$. Total momentum $P = p_1 + p_2$; differentiate: $dP/dt = F_{12} + F_{21} = 0$. So $P = \text{constant}$ (conservation of linear momentum). (c) Momentum: $3 \times 8 + 1 \times 0 = (3 + 1) v \rightarrow 24 = 4 v \rightarrow v = 6 \text{ m/s}$ in the same direction as the original motion.

Standard derivation followed by an inelastic collision computation.

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